Title: Introduction to Java programming and Basic Problem Solving.

Objectives:

- To understand the history and core features of Java
- To understand and explore how a Java program executes (JDK → JRE → JVM)
- To learn basic Java syntax and elementary programming concepts.
- To implement and understand data types, array, loops and simple algorithm.
- To perform linear search and find the maximum element in an array.

Introduction:

Java is a high level, class based, object-oriented Programming language developed by James Gosling in 1995 at Sun Microsystems. It has since become one of the most widely used programming language in the world due to its platform independence, robustness and scalability.

Java follows the principle of "Write once and Run Anywhere", meaning compiled Java code can run on all platforms that support Java without the need for recompilation.

• How Java program runs: The execution of a Java program involves three main components: JDK, JRE and JVM. The entire process can be brake down into the following steps:

1. Writing the code: (.java file)
   - The source code is written using Java syntax and saved with a .java extention.

2. Compilation (using JDK):
   - The Java Development Kit (JDK) includes the Java Compiler (javac) that translates the source code into byte code.
   - Bytecode is stored in a .class file and is platform-independent.

3. Execution (using JRE)
   - The Java Runtime Enviroment (JRE) provides the required libries and tools to run the bytecode.

4. Runtime interpretation (JVM):
   - The Java Virtual Machine (JVM) reads and interprets the byte code, translating it into native machine code using Just-In-Time (JIT) compilation.
   - This allows Java programs to run on any device with JVM installed.

Proposed Methodology:

This lab involves writing a Java program that demonstrates the following elementary concepts:

- Use of premitive data types
- Array initialization and access
- Use of for loop and for each loop
- Linear Search algorithm to find a specific value
- Find the maximum element in an array.

Each section of the code is designed to illustrate how Java handles input / output, control structure, and algorithmic logic:

Program Name: A program print data type in Java

Code:

```
public class dataType {
    public static void main(String[] args){
        byte a = 10;
        short b = 100;
        int c = 100000;
        char d = 'A';
        float e = 3.50f;
        double f = 3.4567;

        System.out.println(a);
        System.out.println(b);
        System.out.println(c);
        System.out.println(d);
        System.out.println(e);
        System.out.println(f);
    }
}
```

Output:

```
C:\Users\abrar\.jdks\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ
10
100
100000
A
3.5
3.4567

Process finished with exit code 0
```

Program Name: A program to access array elements

Code:

```
public class accessArrayElement {
    public static void main(String[] args){
        int[] age = {16, 23, 45};
        System.out.println("Accessing array elements:");
        System.out.println("First Element: " + age[0]);
        System.out.println("Second Element: " + age[1]);
        System.out.println("Third Element: " + age[2]);
    }
}
```

Output:

```
C:\Users\abrar\.jdks\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
Accessing array elements:
First Element: 16
Second Element: 23
Third Element: 45

Process finished with exit code 0
```

Program Name: A program to implement for loop

Code:

```
public class forLoopArray {
    public static void main(String[] args){
        int[] a = {12,23,34,45,56};

        for(int i=0; i<a.length; i++){
            System.out.println("index[" + i +"] = " + a[i]);
        }
    }
}
```

Output:

```
C:\Users\abrar\.jdks\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ
index[0] = 10
index[1] = 20
index[2] = 30
index[3] = 40
index[4] = 50

Process finished with exit code 0
```

Program Name: A program to implement for each loop

Code:

```
public class forEachLoopArray {
    public static void main(String[] args){
        int[] a = {10,20,30,40,50};
        int i = 0;
        for(int x : a){
            System.out.println("index[" + i +"] = " + x);
            i++;
        }
    }
}
```

Output:

```
C:\Users\abrar\.jdks\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ
index[0] = 12
index[1] = 23
index[2] = 34
index[3] = 45
index[4] = 56

Process finished with exit code 0
```

Program Name: A program to search an element in array

Code:

```
public class linearSearch {
    public static void main(String[] args){
        int[] a = {10,20,30,40,50};
        int item = 40;
        int count = 0;
        boolean t = true;

        for(int x : a){
            if(x == item){
                t = false;
                break;
            }
        }
        System.out.println((t) ? "Item Not Found :(" : "Item Found :)");

    }
}
```

Output:

```
C:\Users\abrar\.jdks\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ
Item Found :)

Process finished with exit code 0
```

Program Name: A program to find the maximum value in array

Code:

```
public class maxElement {
    public static void main(String[] args){
        int[] a = {10,20,30,40,50};
        int max = a[0];

        for(int i=1; i<a.length; i++){
            if(max < a[i]){
                max = a[i];
            }
        }
        System.out.println("Maximum Element of the Arrary: " + max);
    }
}
```

Output:

```
C:\Users\abrar\.jdks\openjdk-24.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ
Maximum Element of the Arrary: 50

Process finished with exit code 0
```

Discussion :

In this lab, we explored essential Java programming concepts. The program:

- Demonstrated how to store data in an array.
- Used a for loop to populate the array and a for-each-loop to display the elements.
- Implemented linear search to locate value
- Compared each element of the array to find the maximum value.

This hands on approach provided a solid foundation in both syntax and logic, reinforcing Java's structure and flexibility.

Conclusion :

This lab introduced the Java Programming language its compilation and execution process, and core programming constructors. By implementing a simple program with data input, searching, and comparison, we gained practical knowledge of :

- Java's platfrom independence and execution flow.
- Use of arrays and loops.
- Basic algorithm design and problem-solving.

Mastery of these concepts in crucial for progressing to more advanced topics like object-oriented programming and data structure.

Reference :

01. Oracle Java documentation

02. GeekforGeeks - Java Programming Basic

03. W3Schools Java Tutorial

04. Lecture notes